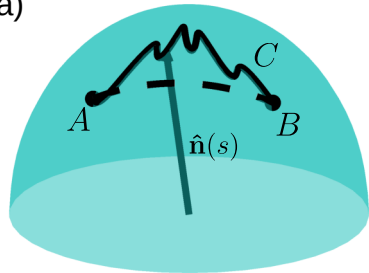


(a)



$$\int_C |\hat{\mathbf{n}} \times d\hat{\mathbf{n}}| = C \text{ arclength}$$

$$\int_C \hat{\mathbf{E}} \cdot (\hat{\mathbf{n}} \times d\hat{\mathbf{n}}) = AB \text{ distance}$$

(b)



$$\int_C \hat{\mathbf{E}} \cdot (\hat{\mathbf{n}} \times d\hat{\mathbf{n}}) = 0$$

(c)



$$\int_C \hat{\mathbf{E}} \cdot (\hat{\mathbf{n}} \times d\hat{\mathbf{n}}) = \pi$$